

Project I : Menu system →

1) Required Components →

- ① Arduino R3
- ② Bread board
- ③ Jumper wire
- ④ 16 x 2 LCD screen
- ⑤ RGB LED
- ⑥ Potentiometer
- ⑦ Push button

② Difficulties →

- ① Making scroll on LCD
- ② Make LED, LCD, Potentiometer, buttonwork simultaneously
- ③ Making animation smooth
- ④ Refresh cycle reduction & No flickering

③ Pin to Pin Connection →

① VCC & Ground → Common to all & on board

② RGB LED →

- Common Cathode → GND
- Red → Pin D11 ~
- Green → Pin D10 ~
- Blue → Pin D9 ~

③ Button → Pull up Resistor

0 = ON , 1 = OFF

Need to Fix this

- VCC & GND → ON Board
- Input → D8

④ Potentiometer →

- VCC & Ground → Common & ON Board
- Variable → Pin 2 to Pin A0

⑤ LCD screen → I2C Protocol

- VCC & GND → Common & ON Board 5V
- SDA → A4 & SCL → A5
- Library → LiquidCrystal — I2C & Wire
↓
To use function LCD
↓
For I2C Protocol.

⑥ Improvement which can be added →

- ① Increase No. of option
- ② Increase code efficiency
- ③ Use 2 more button to move menu up & Down
- ④ Virtual PCB can be made

⑤ What Learn't →

- ① How to debug code
- ② Doing good documentation
- ③ Making & using array
- ④ Do own calculation
- ⑤ keep circuit clean
- ⑥ Scroll working & making own UI
- ⑦ Use pullup res perfectly